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System and method for applying a shear test to a threaded section of a bolt

Abstract

A system having: a shear die set having a fork that includes first and second legs that are have grooves and are spaced from each other to define a test cavity; and a center sleeve that extends between the legs and is radially larger than the first leg groove, the center sleeve has an inner threaded surface and receives a threaded portion of the bolt; a first support sleeve having an inner threaded surface that receives a threaded portion of the bolt, and the first support sleeve has an outer diameter surface that fits in the first leg groove; and a second support sleeve having a same size and shape as the first support sleeve and receives a threaded portion of the bolt and fits in the second leg groove; and the outer diameter of the first support sleeve is different than the outer diameter of the center sleeve.

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Background/Summary

BACKGROUND

(1) The embodiments are directed to shear tests and more specifically to a system and method for applying a shear test to a threaded section of a bolt.

(2) Shear tests are mechanical tests that measure a material's shear strength. In a double shear test, a cylindrical test specimen is placed in a set of dies and compressed in such a way to create two planes of shear that act on a cylinder of the specimen. Typically a set of double shear dies is utilized to compress an unsupported region of a test piece in a three-point-bend jig. This configuration provides two planes of shear with minimal bending. The response of the material is measured, and the results are utilized to characterize the engineering qualities of the material. Standard practices exist, such as NASM1312-13, which is a standard published by the Aerospace Industries Association (www.aia-aerospace.org), directed to fastener test methods for double shear

tests. The standard addresses tests for smooth shanks of bolts, and smooth shafts and cylinders.

BRIEF SUMMARY

(3) Disclosed is a system including: a shear die set having a fork that includes: a fork-base; a first leg extending from the fork-base to a first end; a second leg extending from the fork-base to a second end, wherein: the first and second legs are spaced apart from each other by a first distance to define a test cavity; and the first leg has a first leg groove, and the second leg has a second leg groove that is aligned with the first leg groove and has a same size and shape as the first leg groove; and a center sleeve having an axial length that corresponds with the first distance and is radially larger than the first leg groove, wherein the center sleeve has an inner threaded surface and is configured to receive a center portion of the threaded section of the bolt during the shear test; a first support sleeve having an inner threaded surface that is configured to receive a first portion of the threaded section of the bolt, and the first support sleeve has an outer diameter surface that is sized to fit in the first leg groove during the shear test; and a second support sleeve having a same size and shape as the first support sleeve, wherein the second support sleeve is configured to receive a second portion of the threaded section of the bolt and fit in the second leg groove during the shear test; and the outer diameter surface of the first support sleeve is radially different than the outer diameter surface of the center sleeve.

(4) In addition to one or more of the above disclosed aspects of the system or as an alternate, the first leg groove is defined by a semi-circular surface; and the center sleeve has an outer diameter surface that is radially larger than the semi-circular surface of the first leg groove.

(5) In addition to one or more of the above disclosed aspects of the system or as an alternate, the outer diameter surface of the first support sleeve is radially smaller than the outer diameter surface of the center sleeve.

(6) In addition to one or more of the above disclosed aspects of the system or as an alternate, the system includes a cutter block having a thickness that corresponds with the first distance, the cutter block having a cutter surface with a cutter groove that is sized to receive the center sleeve during the shear test.

(7) In addition to one or more of the above disclosed aspects of the system or as an alternate, the cutter groove has a linear channel portion and a sleeve seat portion, wherein the linear channel portion extends away from the cutter surface to the sleeve seat portion.

(8) In addition to one or more of the above disclosed aspects of the system or as an alternate, the sleeve seat portion of the cutter groove has an inner diameter surface that is sized to correspond with the outer diameter surface of the center sleeve.

(9) In addition to one or more of the above disclosed aspects of the system or as an alternate, the first leg groove and the cutter groove includes smooth surfaces.

(10) In addition to one or more of the above disclosed aspects of the system or as an alternate, the outer diameter surfaces of the center sleeve and the first support sleeve are smooth.

(11) In addition to one or more of the above disclosed aspects of the system or as an alternate, the system includes a base-member that receives and supports the fork and receives and portions of the bolt that extend axially beyond the fork during a shear test.

(12) In addition to one or more of the above disclosed aspects of the system or as an alternate, the base-member includes upstanding legs that include groves for supporting portions of the bolt that extend axially beyond the fork during a shear test.

(13) Further disclosed is a method of shear testing a threaded section of a bolt, including: threading at least a center sleeve around a center portion of the threaded section of the bolt; threading a first support sleeve around a first portion of the threaded section of the bolt, the first portion being located between the center portion and a head of the bolt, so that the first support sleeve and the center sleeve are adjacent to each other; and threading a second support sleeve around a second portion of the threaded section of the bolt, the second portion being between the center portion and end of the bolt, so that the second support sleeve and the center sleeve are adjacent to each other,

wherein an outer diameter surface of the first support sleeve is radially different than an outer diameter surface of the center sleeve; positioning the first support sleeve in a first leg groove of a first leg that extends from fork-base of a fork of a shear die set; positioning the second support sleeve in a second leg groove of a second leg that extends from of the fork-base of the fork, wherein the second leg groove has a same size and shape as the first leg groove, the first and second legs are spaced apart by a first distance to define a test cavity, and the center sleeve has an axial length that corresponds with the first distance and is radially larger than the first leg groove; and applying a cutter block against the center sleeve.

(14) In addition to one or more of the above disclosed aspects of the method or as an alternate, the method includes supporting the fork and portions of the bolt that extend axially beyond the fork with a base-member.

(15) In addition to one or more of the above disclosed aspects of the method or as an alternate, applying the cutter block includes positioning a cutter groove of the cutter block against the center sleeve.

(16) In addition to one or more of the above disclosed aspects of the method or as an alternate, positioning the cutter groove of the cutter block against the center sleeve includes moving the center sleeve through a linear channel portion of the cutter groove until the center sleeve is positioned against a sleeve seat portion of the cutter groove.

(17) Further disclosed is a system for performing a double shear test of a threaded section of a cylinder or bolt, including: a shear die set having a fork and a cutter block; first and second support sleeves that are configured to engage the threaded section of the bolt, wherein the first and second support sleeves are supported by the fork during the shear test; the second support sleeve has a same size and shape as the first support sleeve; a center sleeve that is radially larger than the first and second support sleeves, wherein the center sleeve is configured to engage the threaded section of the bolt, between the first and second support sleeves; and an outer diameter surface of the first support sleeve is radially different than an outer diameter surface of the center sleeve.

(18) In addition to one or more of the above disclosed aspects of the system or as an alternate, the fork has first and second legs spaced apart from each other by a first distance and the center sleeve has an axial length that corresponds with the first distance to provide a slip fit between the center sleeve and the first and second legs of the fork.

(19) In addition to one or more of the above disclosed aspects of the system or as an alternate, the cutter block has a thickness that corresponds with the first distance to provide a slip fit between the cutter block and the first and second legs of the fork; the cutter block has a cutter surface, that faces a fork-base of the fork during the shear test; the cutter surface has a cutter groove that is sized to receive the center sleeve during the shear test; the cutter groove has a linear channel portion that extends away from the cutter surface to a sleeve seat portion; the sleeve seat portion of the cutter groove has an inner diameter surface that is sized to correspond with an outer diameter surface of the center sleeve; and in operation, the cutter groove of the cutter block receives the center sleeve by moving the center sleeve through the linear channel portion of the cutter groove until the center sleeve is positioned against the sleeve seat portion of the cutter groove.

(20) In addition to one or more of the above disclosed aspects of the system or as an alternate, the outer diameter surfaces of the first and second support sleeves and the center sleeve are smooth and the cutter groove is smooth.

(21) In addition to one or more of the above disclosed aspects of the system or as an alternate, the sleeve seat portion of the cutter groove has a semi-circular cross section.

(22) In addition to one or more of the above disclosed aspects of the system or as an alternate, the system includes a base-member that supports the fork and portions of the bolt that extend axially beyond the fork.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an exploded view of a system for shear testing a threaded section of a bolt according to an embodiment, including a fork, cutter block, sleeves and a bolt;
- (2) FIG. 2 is a perspective view of the system;
- (3) FIG. 3 is a detail of the embodiments identified in FIG. 2, showing aspects of the fork and sleeves;
- (4) FIG. 4 is an assembly view of the system, without showing the cutter block;
- (5) FIG. 5 is a further detail of the embodiments identified in FIG. 2, showing aspects of the cutter block;
- (6) FIG. 6 is shows the testing of the bolt with the system;
- (7) FIG. 7 shows additional components of the system in an exploded view;
- (8) FIG. 8 shows the additional components of the system in an assembly view; and
- (9) FIG. 9 is a flowchart showing a method of testing a threaded section of a bolt.

DETAILED DESCRIPTION

(10) Turning to FIGS. 1 and 2, a system **100** is shown for double shear testing of a threaded section **104** of a cylinder or bolt **105**. The system **100** includes a shear die set **110** having a fork **120** and a cutter block **130**. The system **100** includes first and second support sleeves **150**, **160** that engage the threaded section of the bolt **105** are supported by the fork **120** as discussed below. The second support sleeve **160** has a same size and shape as the first support sleeve **150**. A center sleeve **170** that is radially larger than the first and second support sleeves **150**, **160** engages the threaded section of the bolt **105**, between the support sleeves **150**, **160**.

(11) FIGS. 3 and 4 show the fork **120**, the sleeves **150-170** and the bolt **105**. The fork **120** includes a fork-base **200** and a first leg **210** extending from the fork-base **200** to a first end **220** (FIG. 3). A second leg **230** extends from the fork-base **200** to a second end **240** (FIG. 3). The first and second legs **210**, **230** are generally parallel with each other and spaced apart from each other by a first distance **D1** to define a test cavity **250**.

(12) The first leg **210** has a first leg groove **260** formed into its first end **220**. The first leg groove **260** may be defined by a semi-circular surface **265**, and the first leg groove **260** may be smooth. The second leg **230** has a second leg groove **270** formed into its second end **240** that is aligned with the first leg groove **260** and has a same size and shape as the first leg groove **260**. The first and second leg grooves **260**, **270** are configured to support first and second portions **280**, **290** (FIG. 4) of the threaded section **104** of the **105** bolt during a shear test.

(13) The center sleeve **170** has an axial length that corresponds with, e.g., is the same as, the first distance **D1** to provide a slip fit between the center sleeve **170** and the first and second legs **210**, **230** of the fork **120**. The center sleeve **170** is radially larger than the first leg groove **260**. The center sleeve **170** has an inner threaded surface **310** and is configured to receive a center portion **320** (FIG. 4) of the threaded section **104** of the bolt **105** during the shear test. The first portion **280** of the threaded section **104** of the bolt **105** is between the center portion **320** of the threaded section **104** of the bolt **105** and a head **106** of the bolt **105**. With this configuration, the first support sleeve **150** and the center sleeve **170** are axially adjacent to each other during the shear test. The second portion **290** of the threaded section **104** of the bolt **105** is between the center portion **320** and the end **107** of the bolt **105**. With this configuration, the second support sleeve **160** and the center sleeve **170** are adjacent to each other during the shear test.

(14) The first support sleeve **150** has an inner threaded surface **325** and is configured to receive the first portion **280** of the threaded section **104** of the bolt **105**. The second support sleeve **160** is configured to receive the second portion **290** of the threaded section **104** of the bolt **105** during the shear test.

(15) The center sleeve **170** has an outer diameter surface **330** that is radially larger than the semi-circular surface **265** of the first leg groove **260**. An outer diameter surface **340** of the first support sleeve **150** is radially smaller than the outer diameter surface **330** of the center sleeve **170**. The outer diameter surface **340** is sized to fit in the first leg groove **260** during the shear test. That is, the center sleeve **170** has an outer diameter that is oversized to allow for proper alignment of the sleeve **170** between the first and second legs **210**, **230**, e.g., preventing slippage along the leg grooves **260**, **270**. The outer diameter surfaces **330**, **340** of the center sleeve **170** and the first support sleeve **150** are smooth.

(16) It is to be appreciated that reference to the center portion **320** of the threaded section **104** of the bolt **105** is for convenience and not intended to require the shear test be applied to a point halfway along the threaded section **104** of the bolt **105**. Any location intermediate the head **106** and the end **107** of the bolt **105** that can be threaded into the center sleeve **170**, and is axially surrounded by portions **280**, **290** that can be threaded into the support sleeves **150**, **160**, is part of the center portion **320** of the bolt **105**. In addition, the oversized center portion/sleeve **170** aligns the shear plane with the interface of the two sleeves **150**, **160**, which allows for load to only be applied to the test piece (bolt **105**), and not the sleeves **150-170**. This improves data quality, and is easier to set up.

(17) Turning to FIG. 5, the cutter block **130** has a thickness that corresponds with, e.g., is equal to, the first distance **D1** to provide a slip fit between the cutter block **130** and the first and second legs **210**, **230** of the fork **120**. The cutter block **130** has a cutter surface **400**, that faces the fork-base **200** of the fork **120** when the cutter block **130** is used during a shear test. The cutter surface **400** has a cutter groove **410** that is sized to receive the center sleeve **170** during the shear test. The cutter groove **410** may be smooth. The cutter groove **410** has a linear channel portion **420** that extends away from the cutter surface **400** to a sleeve seat portion **430**. The sleeve seat portion **430** of the cutter groove **410** has an inner diameter surface **440** that is sized to correspond with, e.g., is the same size as, the outer diameter surface **330** of the center sleeve **170**. That is, the sleeve seat portion **430** has a semi-circular cross section. The cutter groove **410** of the cutter block **130** receives the center sleeve **170** by moving the center sleeve **170** through the linear channel portion **420** of the cutter groove **410** until the center sleeve **170** is positioned against the sleeve seat portion **430** of the cutter groove **410**. The cutter surface **400** has chamfered edges **450**, **460** to enable orienting the cutter block **130** in the test cavity **250** during a shear test.

(18) Turning to FIG. 6, the disclosed embodiments impose a shear force on the specimen, e.g., the bolt **105**, and reduces or prevents minimal bending forces from developing along the center portion **320** (FIG. 4) of the bolt **105** within the test cavity **250**. The embodiments provide for the double shear testing of the threaded section **104** of the bolt **105** by the inclusion of the outer support sleeves **150**, **160** for supporting the bolt **105**, and the oversized center sleeve **170**. The sleeves **150-170** are threaded tubes and the center sleeve **170** is sized so its axial ends are on the planes of shear, which are aligned with first and second inner surfaces **500**, **510** of the first and second legs **210**, **230** of the fork **120**. The center sleeve **170** does not contribute to carrying shear load for the bolt **105**. The oversized center sleeve **170** which allows for proper alignment of the sleeve **170** between the first and second legs **210**, **230** provides for accurate measurements. The cutter block **130** has the oversized cutter groove **410** that is sized to receive and seat the center sleeve **170**. With the disclosed embodiments, threads of the bolt **105** are supported by the outer sleeves **150**, **160**, and all sleeves **150-170** have smooth outer surfaces, to prevent damage to the shear die **110** and prevent skewing shear test results that could result from the deformation of the threads.

(19) Turning to FIGS. 7 and 8, the system **100** is a die assembly that also includes a base-member **500** that seats therein the fork **120**. The base-member **500** has a base **510** and upstanding legs **520** and **530** that support the fork-base **200**, fork legs **210**, **230**, the head **106** of the bolt **105** and the end **107** of the bolt **105** in grooves **540**, **550**. The grooves **540**, **550** in the base-member legs **520**, **540** are configured similarly to the grooves **260**, **270** in the fork **120**. The base-member **500** provides

support while the cutter block **130** is applied during a double shear test. During this time, the sleeves **150-170** support the bolt **105** as indicated above, while the base-member **500** that help braces the fork **120** to prevent bending in the system **100**.

(20) Turning to FIG. **9**, a flowchart shows a method of shear testing the threaded section **104** of the bolt **105**. As shown in block **10**, the method includes threading the center sleeve **170** around the center portion **320** of the threaded section **104** of the bolt **105**. As shown in block **10A**, the method includes block **10A** of threading the first support sleeve **150** around the first portion **280** of the threaded section **104** of the bolt **105**. As shown in block **10B**, the method includes threading the second support sleeve **160** around the second portion **290** of the threaded section **104** of the bolt **105**.

(21) As shown in block **20A**, the method includes positioning the first support sleeve **150** in the first leg groove **260**. As shown in block **20B**, the method **270** includes positioning the second support sleeve **160** in the second leg groove **270**.

(22) As shown in block **30**, the method includes utilizing a base-member **500** to support the fork **120** and portions of the bolt **105** that extend beyond the fork **120**. As shown in block **40**, the method includes applying the cutter block **130** against the center sleeve **170**. As shown in block **40A**, applying the cutter block **130** includes positioning the cutter groove **410** of the cutter block **130** against the center sleeve **170**. As shown in block **40A1**, this includes moving the center sleeve **170** through the linear channel portion **420** of the cutter groove **410** until the center sleeve **170** is positioned against the sleeve seat portion **430** of the cutter groove **410**.

(23) The embodiments allow for shear data to be generated for threaded section **104** of the bolt **105** and other threaded fasteners, cylinders and specimen. This allows for the empirical validation and understanding of fasteners and threaded components, as well as for the improvement of material models that this data supports. This also allows for the understanding and realization through mechanical testing of the benefits of specialized processing techniques of fasteners that otherwise may be difficult to model. The embodiments may be applied to the development of testing specifications and quality controls around fasteners. For example, fasteners that include cold-rolled threads for strengthening effects can be testing in shear to measure the impact of the cold-rolled processing method. The center oversized sleeve **170** of the embodiments allow for relatively easy alignment of the sleeves and result in more accurate shear test data.

(24) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(25) Those of skill in the art will appreciate that various example embodiments are shown and described herein, each having certain features in the particular embodiments, but the present disclosure is not thus limited. Rather, the present disclosure can be modified to incorporate any number of variations, alterations, substitutions, combinations, sub-combinations, or equivalent arrangements not heretofore described, but which are commensurate with the scope of the present disclosure. Additionally, while various embodiments of the present disclosure have been described, it is to be understood that aspects of the present disclosure may include only some of the described embodiments. Accordingly, the present disclosure is not to be seen as limited by the foregoing description, but is only limited by the scope of the appended claims.

Claims

1. A system comprising: a shear die set having a fork that includes: a fork-base; a first leg extending from the fork-base to a first end; a second leg extending from the fork-base to a second end, wherein: the first and second legs are spaced apart from each other by a first distance to define a test cavity; and the first leg has a first leg groove, and the second leg has a second leg groove that is aligned with the first leg groove and has a same size and shape as the first leg groove; and a center sleeve having an axial length that corresponds with the first distance and is radially larger than the first leg groove, wherein the center sleeve has an inner threaded surface and is configured to receive a center portion of the threaded section of the bolt during the shear test; a first support sleeve having an inner threaded surface that is configured to receive a first portion of the threaded section of the bolt, and the first support sleeve has an outer diameter surface that is sized to fit in the first leg groove during the shear test; and a second support sleeve having a same size and shape as the first support sleeve, wherein the second support sleeve is configured to receive a second portion of the threaded section of the bolt and fit in the second leg groove during the shear test; and the outer diameter surface of the first support sleeve is radially different than the outer diameter surface of the center sleeve.
2. The system of claim 1, wherein: the first leg groove is defined by a semi-circular surface; and the center sleeve has an outer diameter surface that is radially larger than the semi-circular surface of the first leg groove.
3. The system of claim 2, wherein: the outer diameter surface of the first support sleeve is radially smaller than the outer diameter surface of the center sleeve.
4. The system of claim 3, comprising: a cutter block having a thickness that corresponds with the first distance, the cutter block having a cutter surface with a cutter groove that is sized to receive the center sleeve during the shear test.
5. The system of claim 4, wherein: the cutter groove has a linear channel portion and a sleeve seat portion, wherein the linear channel portion extends away from the cutter surface to the sleeve seat portion.
6. The system of claim 5, wherein: the sleeve seat portion of the cutter groove has an inner diameter surface that is sized to correspond with the outer diameter surface of the center sleeve.
7. The system of claim 6, wherein: the first leg groove and the cutter groove comprise smooth surfaces.
8. The system of claim 7, wherein: the outer diameter surfaces of the center sleeve and the first support sleeve are smooth.
9. The system of claim 1, including a base-member that receives and supports the fork and receives and portions of the bolt that extend axially beyond the fork during a shear test.
10. The system of claim 9, wherein the base-member includes upstanding legs that include grooves for supporting portions of the bolt that extend axially beyond the fork during a shear test.
11. A method of shear testing a threaded section of a bolt, comprising: threading at least a center sleeve around a center portion of the threaded section of the bolt; threading a first support sleeve around a first portion of the threaded section of the bolt, the first portion being located between the center portion and a head of the bolt, so that the first support sleeve and the center sleeve are adjacent to each other; and threading a second support sleeve around a second portion of the threaded section of the bolt, the second portion being between the center portion and end of the bolt, so that the second support sleeve and the center sleeve are adjacent to each other; wherein an outer diameter surface of the first support sleeve is radially different than an outer diameter surface of the center sleeve; positioning the first support sleeve in a first leg groove of a first leg that extends from fork-base of a fork of a shear die set; positioning the second support sleeve in a second leg groove of a second leg that extends from of the fork-base of the fork, wherein the second leg groove has a same size and shape as the first leg groove, the first and second legs are spaced apart by a first distance to define a test cavity, and the center sleeve has an axial length that

corresponds with the first distance and is radially larger than the first leg groove; and applying a cutter block against the center sleeve.

12. The method of claim 11, including supporting the fork and portions of the bolt that extend axially beyond the fork with a base-member.

13. The method of claim 12, wherein applying the cutter block includes positioning a cutter groove of the cutter block against the center sleeve.

14. The method of claim 13, wherein positioning the cutter groove of the cutter block against the center sleeve includes moving the center sleeve through a linear channel portion of the cutter groove until the center sleeve is positioned against a sleeve seat portion of the cutter groove.

15. A system for performing a double shear test of a threaded section of a cylinder or bolt, comprising: a shear die set having a fork and a cutter block; first and second support sleeves that are configured to engage the threaded section of the bolt, wherein the first and second support sleeves are supported by the fork during the shear test; the second support sleeve has a same size and shape as the first support sleeve; a center sleeve that is radially larger than the first and second support sleeves, wherein the center sleeve is configured to engage the threaded section of the bolt, between the first and second support sleeves; and an outer diameter surface of the first support sleeve is radially different than an outer diameter surface of the center sleeve.

16. The system of claim 15, wherein: the fork has first and second legs spaced apart from each other by a first distance and the center sleeve has an axial length that corresponds with the first distance to provide a slip fit between the center sleeve and the first and second legs of the fork.

17. The system of claim 16, wherein: the cutter block has a thickness that corresponds with the first distance to provide a slip fit between the cutter block and the first and second legs of the fork; the cutter block has a cutter surface, that faces a fork-base of the fork during the shear test; the cutter surface has a cutter groove that is sized to receive the center sleeve during the shear test; the cutter groove has a linear channel portion that extends away from the cutter surface to a sleeve seat portion; the sleeve seat portion of the cutter groove has an inner diameter surface that is sized to correspond with an outer diameter surface of the center sleeve; and in operation, the cutter groove of the cutter block receives the center sleeve by moving the center sleeve through the linear channel portion of the cutter groove until the center sleeve is positioned against the sleeve seat portion of the cutter groove.

18. The system of claim 17, wherein: the outer diameter surfaces of the first and second support sleeves and the center sleeve are smooth and the cutter groove is smooth.

19. The system of claim 18, wherein the sleeve seat portion of the cutter groove has a semi-circular cross section.

20. The system of claim 15, including a base-member that supports the fork and portions of the bolt that extend axially beyond the fork.
